

BÖLÜM-2 BJT ve JFET FREKANS CEVABI

— Kapasite & Frekans ilişkisi:

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$C = 2\text{pF}$ ise

$f = 1\text{Hz}$ 'de $X_{C\ 1\text{Hz}} = \frac{1}{2\pi \cdot 1 \cdot 2\text{pF}} = \frac{10^{12}}{4\pi} \approx \infty$ (açık devre)

$f = 10\text{MHz}$ 'de $X_{C\ 10\text{MHz}} = \frac{1}{2\pi \cdot 10 \cdot 10^6 \cdot 2 \cdot 10^{-12}} = 7,96\text{k}\Omega$ (ihmal edilemez)

$C = 100\mu\text{F}$ ise

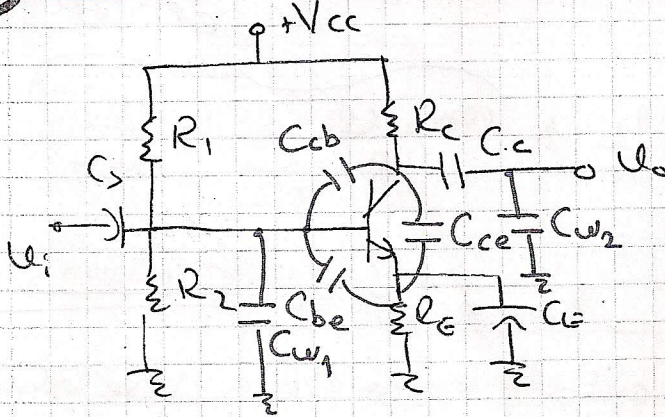
$f = 10\text{MHz}$ 'de $X_{C\ 10\text{MHz}} = \frac{1}{2\pi \cdot 10 \cdot 10^6 \cdot 100 \cdot 10^{-6}} = \frac{1}{2000\pi} \approx 0$ (kısık devre)

$f = 1\text{Hz}$ 'de $X_{C\ 1\text{Hz}} = \frac{1}{2\pi \cdot 1 \cdot 100 \cdot 10^{-6} \cdot 1} = \frac{10^6}{200\pi} = 1,6\text{k}\Omega$ (ihmal edilemez)

NOT: μF 'lor düzeyindeki kapasiteler yüksek frekanslarda kısık devre alınabilir fakat düşük frekanslarda ihmal edilemez.

pF 'lor düzeyindeki kapasiteler ise yüksek frekanslarda ihmal edilemez.

① BJT Frekans Etkisi:

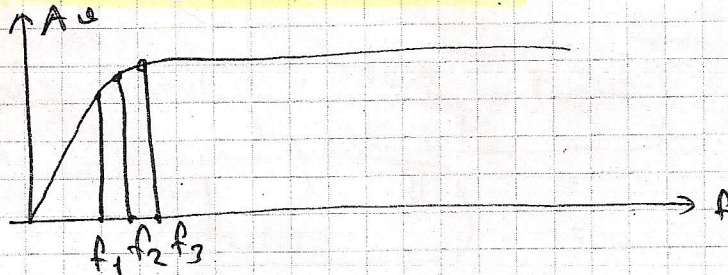


— C_s, C_c, C_E : kutiplama ve köprüleme kondansatörleri. μF 'lor düzeyinde. Orta ve yüksek frekanslarda kısık devre yapılır. Fakat çok düşük frekanslarda (0-100kHz) etkiler değerlendirilir.

— $C_{bc}, C_{ce}, C_{be}, C_{W1}, C_{W2}$: transistörün iç yapısında kaynaklanan kapasiteler. C_{W1} ve C_{W2} : kablo kuma kapasiteleri. pF 'lor düzeyinde. düşük ve orta frekans bölgesinde açık devre yapılır. Fakat yüksek frekanslarda

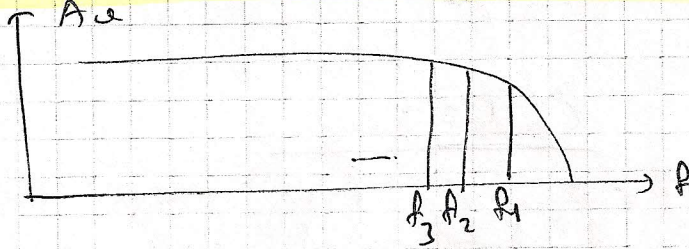
(MHz'ler gibi) etkiler dikkate alınmalıdır.

② C_s, C_c ve C_E kapasiteler düşük frekanslarda yüksek geçiren filtre gibi davranır :

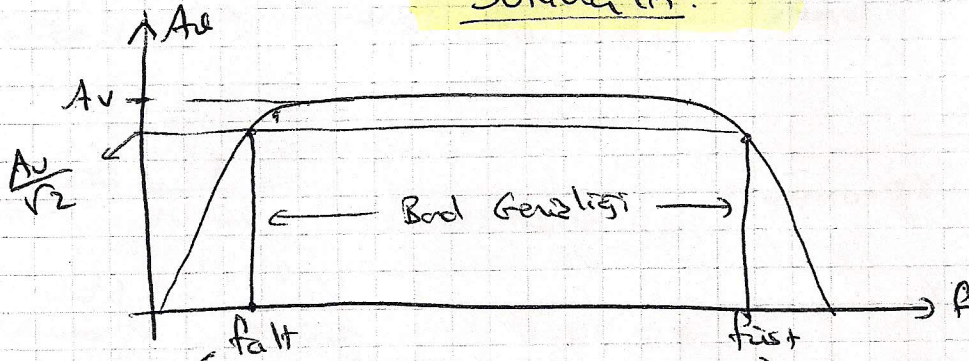


* Yüksek frekanslarda çıkınca transistörün jonsiyonları arasında kapasiteler oluşur.
 C_{bc}
 C_{ce}
 C_{be}

* C_{bc} , C_{ce} , C_{be} , C_{m1} ve C_{m2} kapasiteler de yuksek frekanslarda alçak geciren filtre gibi davranır.



SONUÇTA:



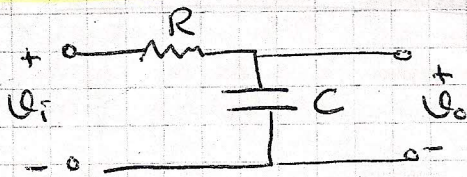
C_s, C_c, C_E etkili

$C_{bc}, C_{ce}, C_{be}, C_{m1}$ ve C_{m2} etkili

Band genişliği = $f_{üst} - f_{alt}$

② Filtre Devreleri:

① Alçak Geçiren Filtre:

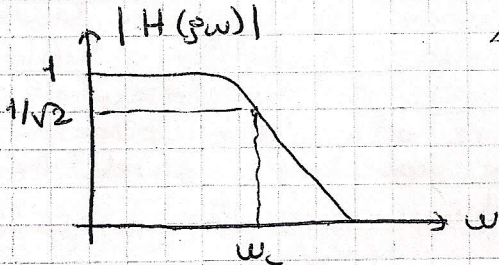


$$H(j\omega) = \frac{U_o}{U_i} \Rightarrow U_o = \frac{U_i}{R + \frac{1}{j\omega C}} \cdot \frac{1}{j\omega C}$$

Gerilim Bölücü

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

transfer fonksiyonu.

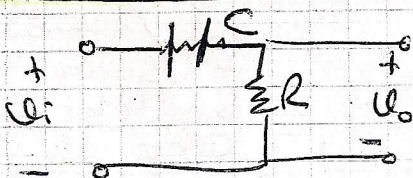


ω_c : cut-off frekansı (kesim, köşe frekansı)

$$\Rightarrow |H(j\omega_c)| = \frac{1}{\sqrt{2}} \Rightarrow \omega_c = 1/RC$$

$$f_c = \frac{1}{2\pi RC}$$

② Yüksek Geçiren Filtre:



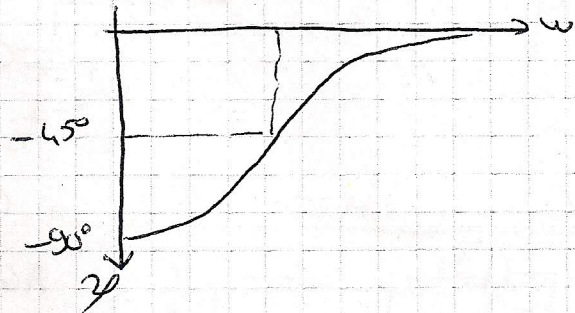
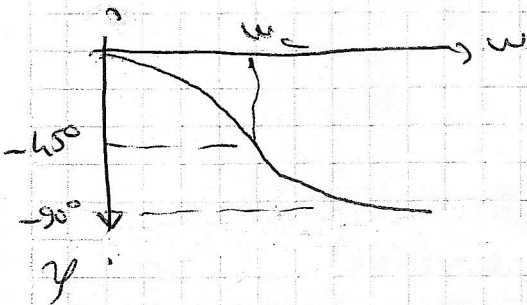
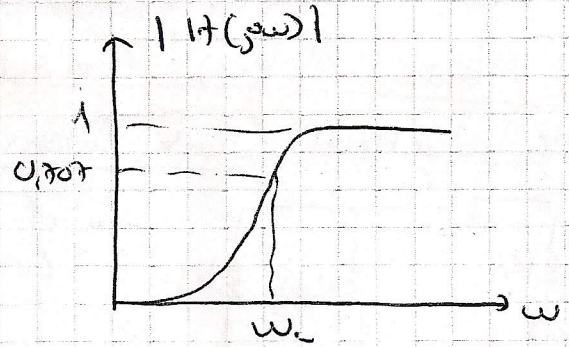
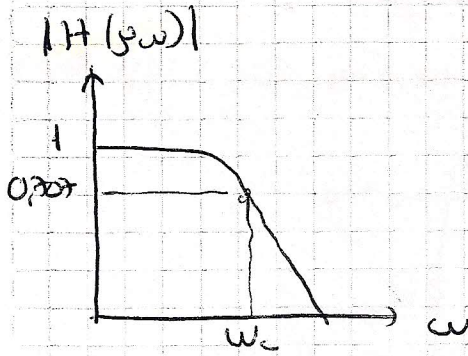
$$H(j\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

transfer fonksiyonu.

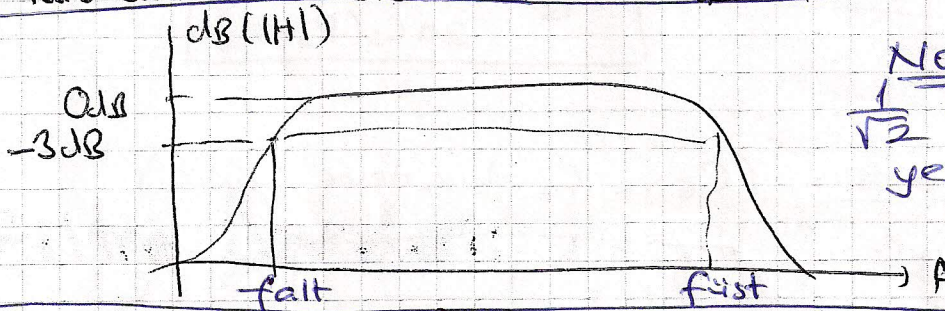
$$f_c = \frac{1}{2\pi RC}$$

Akık Geçiren Filtre

Vütsel Geçiren Filtre

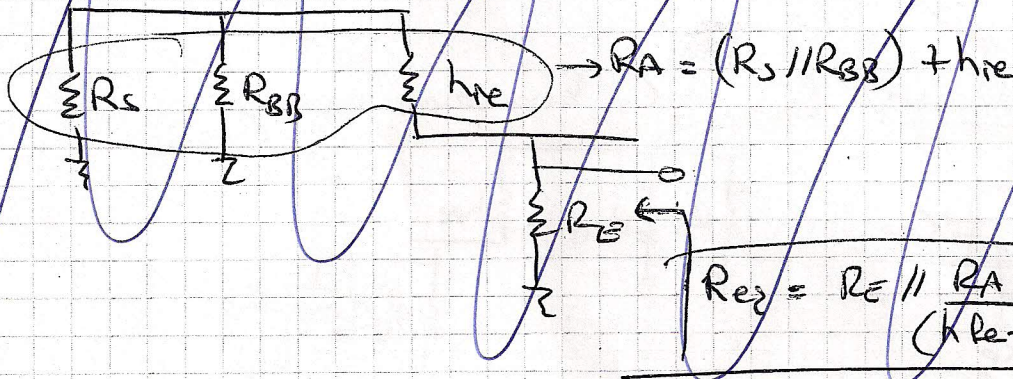


dB türünden transistörün transfer karakteristiği:



NOT: Kazancın $\frac{1}{\sqrt{2}}$ katına düştüğü yerde 3dB kayıp (fark) oluşur.

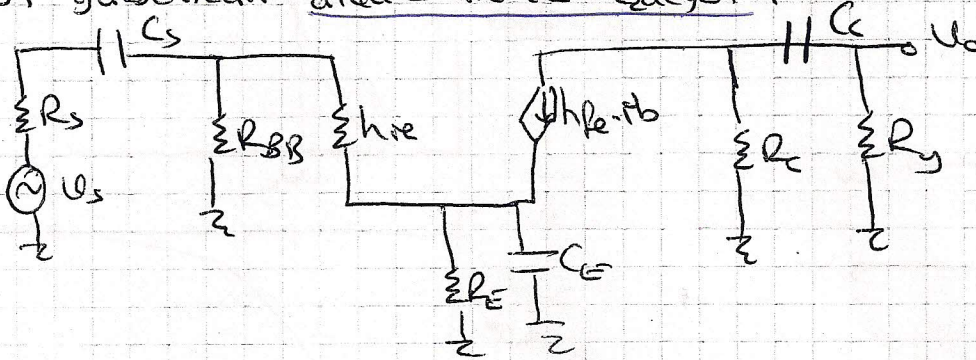
③ R_{eq} (CE K.M.):



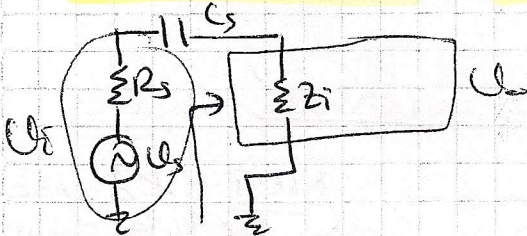
④ BJT'nin frekans Analizi:

① Alçak frekans Analizi: ($C_{bc}, C_{ce}, C_{be}, C_{w1}, C_{w2}$: acık devre)

BJT yükletilen alçak frekans eşdeğeri:



* C_s 'nin etkisi: (C_E ve C_C kısa devre yapılır.)

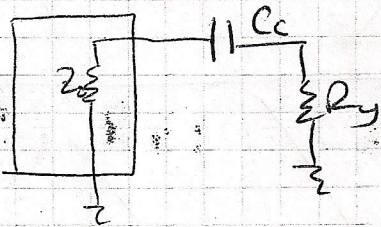


AGF (Alçakgeçiren filtre).

$$f_{1s} = \frac{1}{2\pi C_s (R_s + Z_i)}$$

$$Z_i = R_{BB} \parallel h_{ie}$$

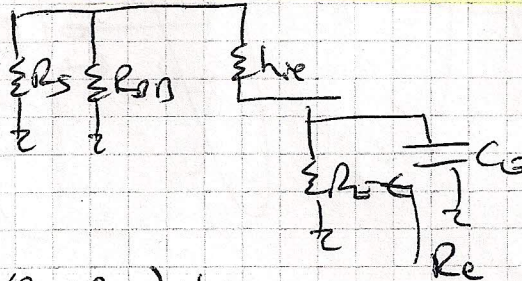
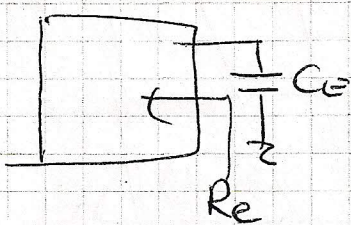
* C_c 'nin etkisi: (C_E ve C_s kısa devre yapılır.)



$$f_{1c} = \frac{1}{2\pi C_c (R_o + Z_o)}$$

$$Z_o = R_C$$

* C_E 'nin etkisi: (C_c ve C_s kısa devre yapılır.)



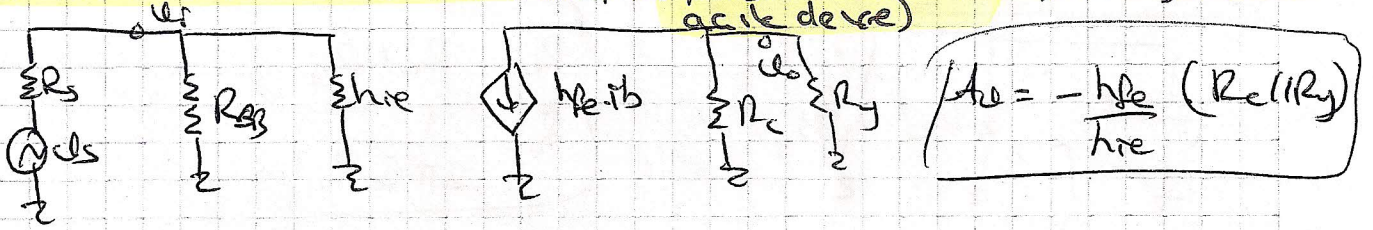
$$R_e = R_E \parallel \frac{R_A}{1 + h_{fe}}$$

$$R_A = (R_s \parallel R_{BB}) + h_{ie}$$

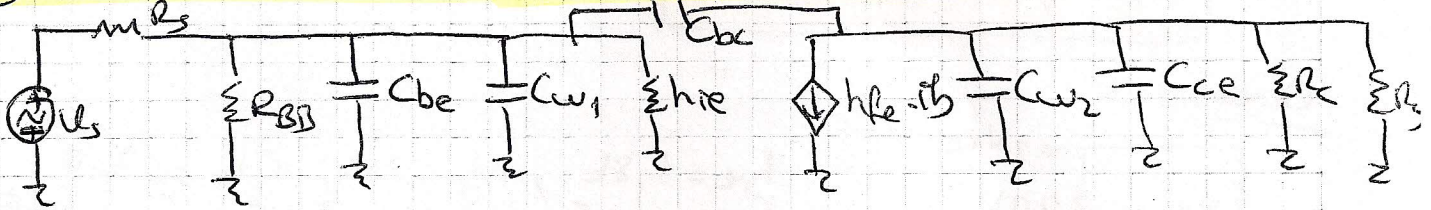
$$f_{1E} = \frac{1}{2\pi C_E \cdot R_e}$$

Bulunan bu üç f_{1s} , f_{1c} ve f_{1E} frekanslarında herhangi yitirilecek de alt kesim frekansı olur. $f_{alt} = \max(f_{1s}, f_{1c}, f_{1E})$

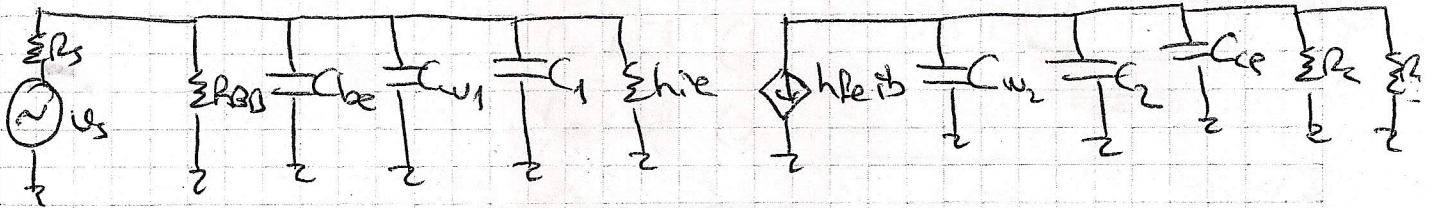
② Orta Frekans Analizi (C_c, C_e, C_s kua devre, diğerler açık devre)



③ Yüksek Frekans Analizi (C_c, C_e, C_s kua devre)

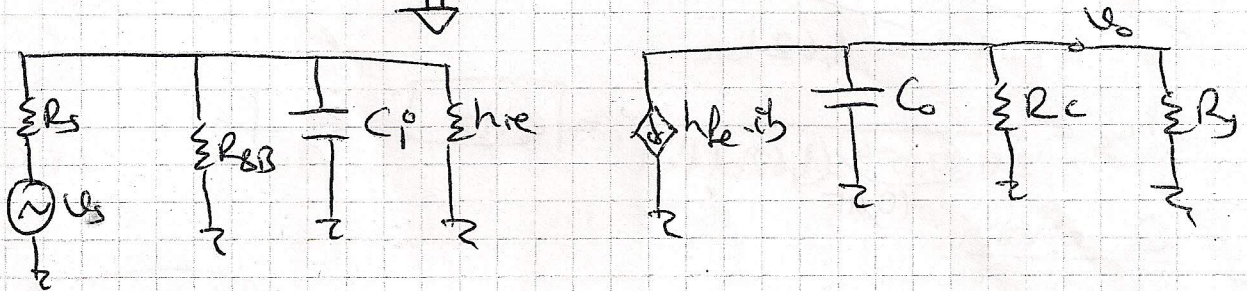


C_{bc} 'nin etkili Miller teorisi ile C_1 ve C_2 şeklinde girip ve çıkışa yansıtılabilir:

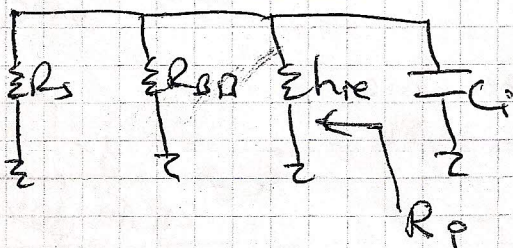


$$C_p = C_{be} + C_{w1} + C_1 \rightarrow C_1 = C_{bc}(1 - A_v)$$

$$C_o = C_{ce} + C_{w2} + C_2 \rightarrow C_2 = C_{bc}(1 - \frac{1}{A_v})$$



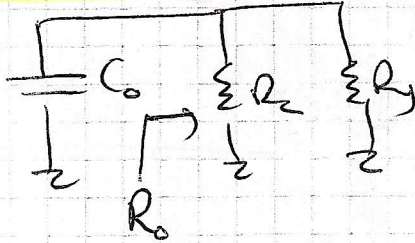
* C_p 'nin etkisi:



$$\Rightarrow R_i = R_s || R_{BB} || h_{ie}$$

$$f_p = \frac{1}{2\pi C_p \cdot R_i}$$

* Colin etkisi:

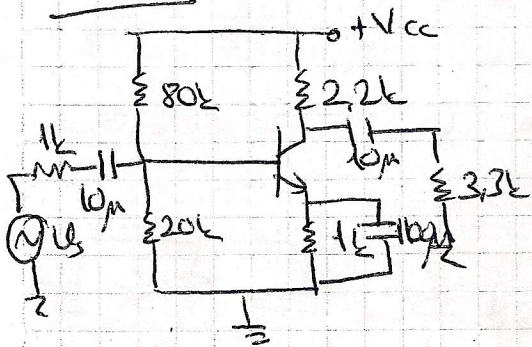


$$R_0 = R_1 || R_2$$

$$f_0 = \frac{1}{2\pi C_0 R_0}$$

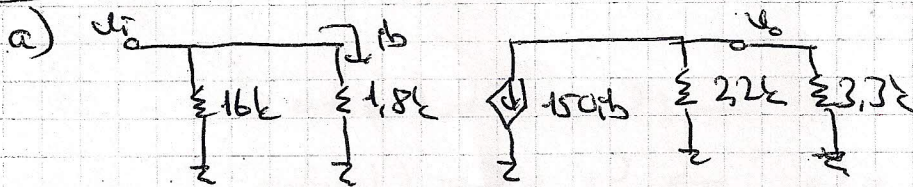
$f_{üst} = \min(f_0, f_0)$

Örnek:



$h_{ie} = 1,8k$, $h_{fe} = 150$, $C_{bc} = 3pF$
 $C_{ce} = 4pF$, $C_{be} = 8pF$, $C_{w1} = 10pF$, $C_{w2} = 10pF$

- a) orta frekans bölgesinde $A_v = ?$
 b) $f_{alt} = ?$
 c) $f_{üst} = ?$



$$A_v = -\frac{150}{1,8k} (2,2k || 3,3k)$$

$$A_v = -110$$

b) C_s etkisi: R_{BB} h_{ie}

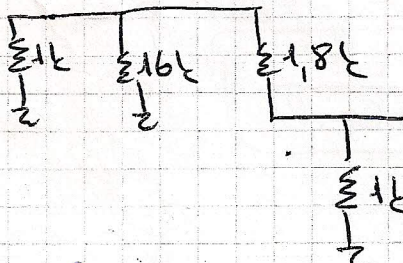
$$Z_i = 16k || 1,8k = 1,62k$$

$$f_{cs} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (1,62k + 1k)} = 61Hz \leftarrow f_{cs} = \frac{1}{2\pi C_s (R_s + Z_i)}$$

C_c etkisi:

$$Z_o = 2,2k \Rightarrow f_{cc} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (2,2k + 3,3k)} \approx 2,9Hz$$

C_E etkisi:



$R_s || R_{BB} + h_{ie}$
 $R_A = (1k || 16k) + 1,8k$

$$R_A = 2,74k$$

$$f_{CE} = \frac{1}{2\pi C_E R_E}$$

$$f_{CE} = \frac{1}{2\pi \cdot 100 \cdot 10^{-6} \cdot 17,82}$$

$$f_{CE} = 89,35Hz$$

$$R_E = \frac{R_A}{1 + h_{fe}} || R_E$$

$$R_E = \frac{R_A}{151} || 1k = 18,5 || 1000$$

$$R_E = 17,82\Omega$$

6

$$f_{alt} = \max(f_{cc}, f_{cs}, f_{ce}) \quad \text{old. 5m}$$

$$f_{alt} = 89,35 \text{ Hz}$$

$$C_1 = C_{bc} (1 - A_v) = 3p \cdot (1 - (-110)) = 333pF$$

$$C_i = C_{be} + C_{w1} + C_1 = 8p + 10p + 333p = 351pF/k$$

$$R_i = R_s // R_{B1} // h_{ie} = 1k // 16k // 1,8k = 0,617k \Omega$$

$$f_i = \frac{1}{2\pi \cdot 351 \cdot 10^{-12} \cdot 0,617k} = 735,2 \text{ kHz} \leftarrow f_i = \frac{1}{2\pi C_i R_i}$$

$$C_2 = C_{bc} \left(1 - \frac{1}{A_v}\right) = 3p \left(1 - \left(\frac{1}{-110}\right)\right) \approx 3pF$$

$$R_o = R_c // R_L$$

$$C_o = C_{ce} + C_{w2} + C_2 = 4p + 10p + 3p = 17pF$$

$$R_o = 33k // 12k = 9,32k$$

$$f_o = \frac{1}{2\pi \cdot 17 \cdot 10^{-12} \cdot 9,32k} \leftarrow f_o = \frac{1}{2\pi C_o R_o}$$

$$f_o = 7,09 \text{ MHz}$$

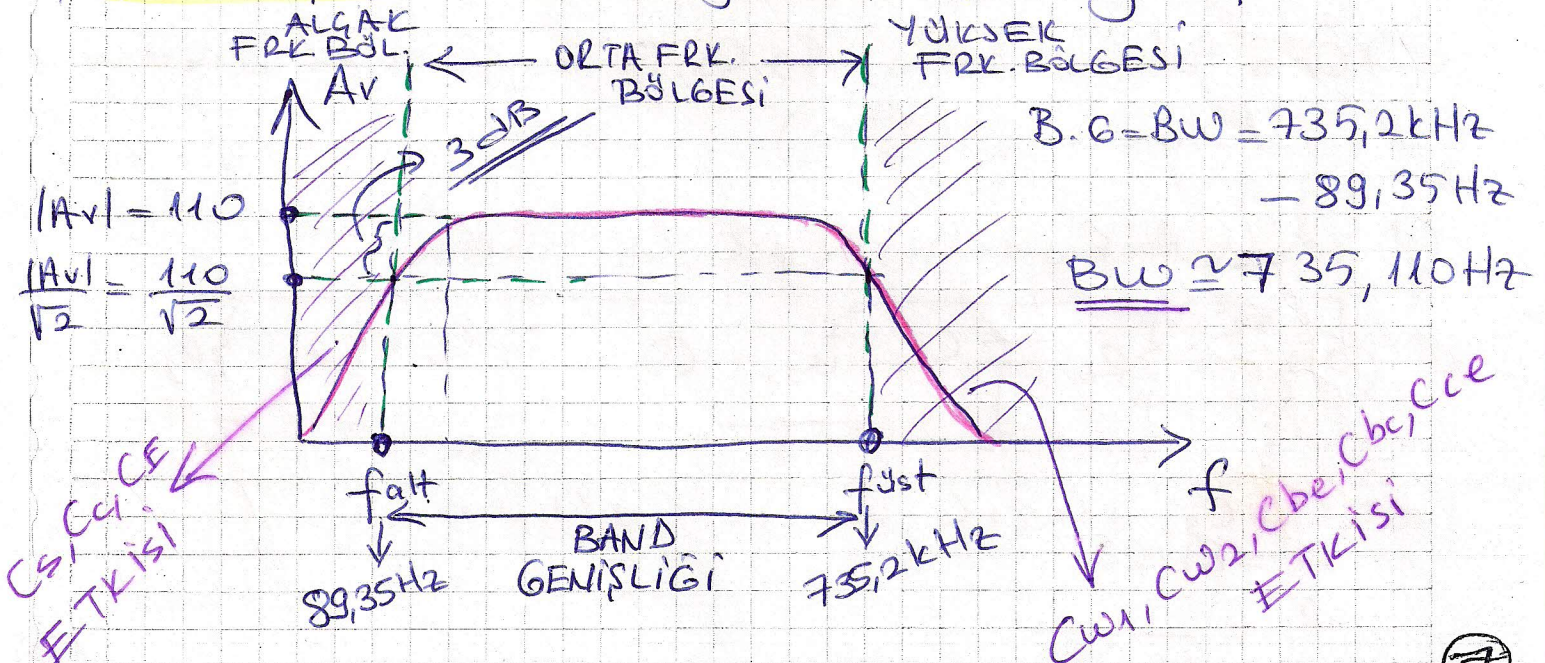
$$f_{üst} = 735,2 \text{ kHz}$$

$$f_{üst} = \min(f_i, f_o)$$

$$f_{üst} = \min(735,2 \text{ kHz}, 7090 \text{ kHz})$$

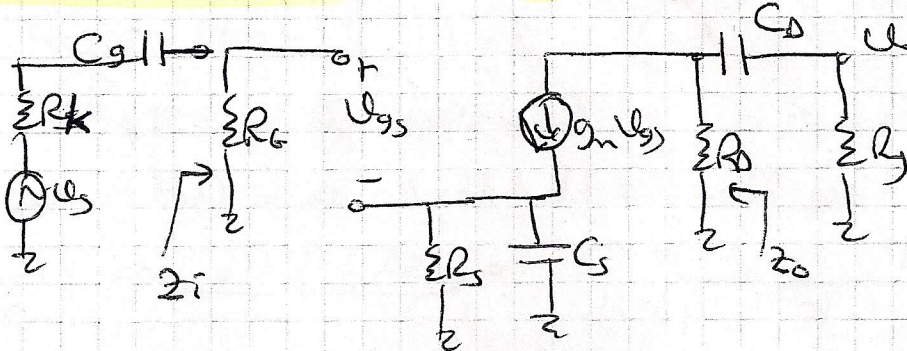
$$f_{üst} = 735,2 \text{ kHz}$$

* Kazanç - frekans Eğrisini Çizseydik!



5. FET Frekans Analizi:

① Alçak frekans analizi: (C_{gs} , C_{ds} , C_{gd} , ω_1 , ω_2 asık devre)



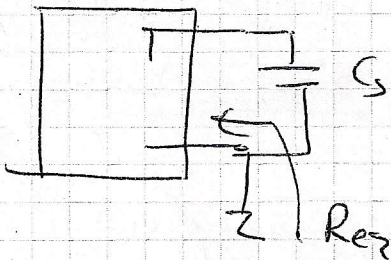
* C_{gs} 'nin etkisi: (C_s ve C_d kuad devre)

$$Z_i = R_g \quad f_{1C_g} = \frac{1}{2\pi(R_g + R_i) \cdot C_g} \angle$$

* C_{ds} 'nin etkisi: (C_s ve C_g kısa devre)

$$Z_o = R_d \quad f_{1C_d} = \frac{1}{2\pi(R_d + R_L) \cdot C_d} \angle$$

* C_s 'nin etkisi: (C_g ve C_d kısa devre)



$$R_{eq} = R_s \parallel \frac{1}{g_m}$$

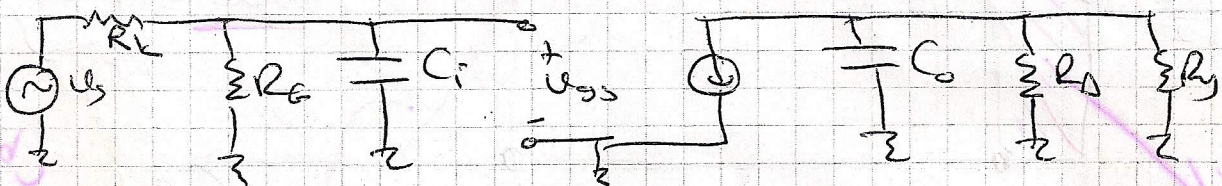
$$f_{1C_s} = \frac{1}{2\pi C_s \cdot R_{eq}}$$

$$f_{alt} = \max(f_{1C_g}, f_{1C_d}, f_{1C_s})$$

② Orta frekans analizi: (C_g , C_s , C_d ku, diğerler asık devre)

$$A_v = -g_m \cdot (R_d \parallel R_L)$$

③ Yüksek frekans analizi: (C_g , C_s , C_d kuad devre)



$$C_i = C_{gs} + C_{w1} + C_1 \rightarrow C_1 = C_{gd}(1 - A_v)$$

$$R_i = R_L \parallel R_g$$

$$f_o = \frac{1}{2\pi C_o R_o}$$

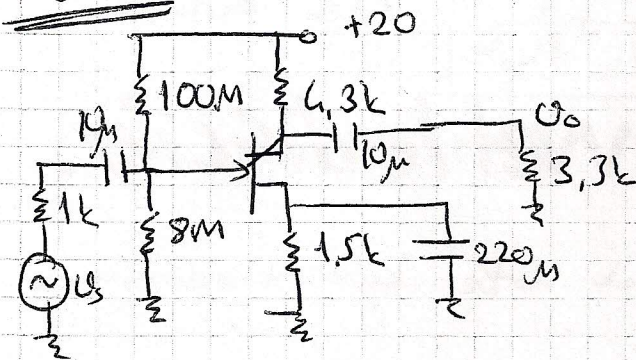
$$C_0 = C_{ds} + C_2 + C_{w2} \rightarrow C_2 = C_{gd} \left(1 - \frac{1}{A_v}\right)$$

$$R_0 = R_D // R_{wy}$$

$$f_0 = \frac{1}{2\pi C_0 R_0}$$

$$f_{\text{cut}} = \min(f_c, f_0)$$

Örnek:



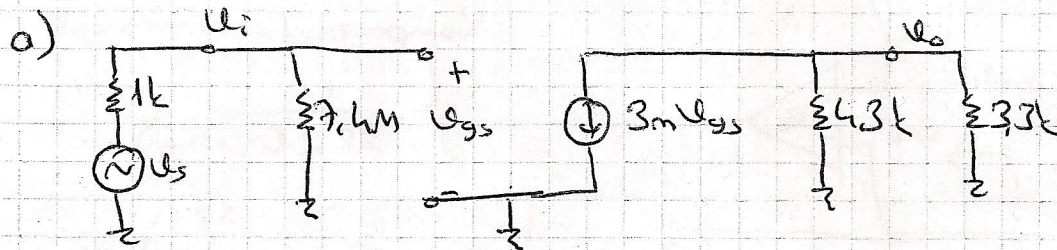
$$g_m = 3 \text{ mS} \quad C_{gs} = C_{ds} = 6 \text{ pF} \quad C_{gd} = 10 \text{ pF}$$

- a) orta frekansa $A_v = ?$
 b) alt kerrn = ?
 c) üst kerrn = ?

$$A_v = -g_m(R_D // R_y)$$

$$A_v = -3 \text{ m} (4.3 \text{ k} // 3.3 \text{ k})$$

$$A_v = -5.6$$



b) $Z_i = 7.4 \text{ M}$, $Z_o = 4.3 \text{ k}$

C_g $f_1 = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} (7.4 \text{ M} // 1 \text{ k})} \approx 0.002 \text{ Hz} // \quad f_{1cg} = \frac{1}{2\pi (Z_i + R_k) \cdot C_g}$

C_D $f_2 = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} (4.3 \text{ k} + 3.3 \text{ k})} \approx 2.09 \text{ Hz} // \quad f_{cD} = \frac{1}{2\pi (R_D + R_y) \cdot C_D}$

C_s $R_{e2} = R_s // \frac{1}{g_m} = 1.5 \text{ k} // \frac{1}{3 \text{ m}} = 1.5 \text{ k} // 0.33 \text{ k} = 0.27 \text{ k}$

$$f_3 = \frac{1}{2\pi \cdot 220 \cdot 10^{-6} \cdot 0.27 \text{ k}} = 2.68 \text{ Hz}$$

$$f_{cs} = \frac{1}{2\pi C_s (R_s // \frac{1}{g_m})}$$

$$f_{\text{alt}} = 2.68 \text{ Hz}$$

c) $C_1 = 10 \text{ p} \cdot (1 - (-5.6)) = 66 \text{ pF}$

$$\left. \begin{array}{l} C_0 = 66 \text{ p} + 20 \text{ p} + 6 \text{ p} = 92 \text{ pF} \\ R_0 = 1 \text{ k} // 7.4 \text{ M} \approx 1 \text{ k} \end{array} \right\} f_0 = \frac{1}{2\pi \cdot 92 \cdot 10^{-12} \cdot 1 \text{ k}} = 1.73 \text{ MHz}$$

$$C_2 = 10 \text{ p} \left(1 - \frac{1}{-5.6}\right) = 11.78 \text{ pF}$$

$$C_0 = 6 \text{ p} + 11.78 \text{ p} + 20 \text{ p} = 37.78 \text{ pF}$$

$$R_0 = 4.3 \text{ k} // 3.3 \text{ k} = 1.86 \text{ k}$$

$$f_0 = \frac{1}{2\pi \cdot 37.78 \cdot 10^{-12} \cdot 1.86 \text{ k}} = 2.26 \text{ MHz}$$

$$f_{\text{cut}} = 1.73 \text{ MHz}$$

(9)

* alt yada üst kesim frekansı bulunırken n adet birbirinden yakın ~~frekans~~ kuşak varsa:

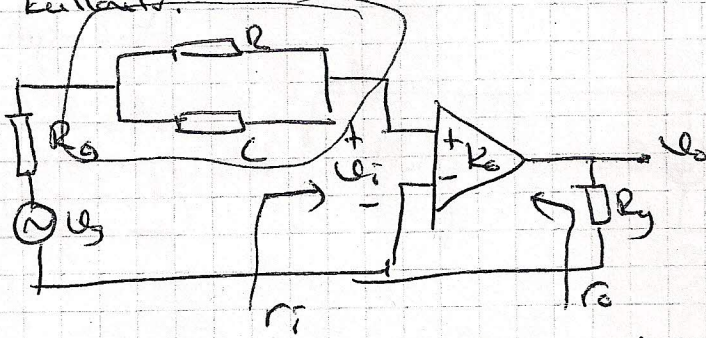
$$f_k = (f_{k1} f_{k2} \dots f_{kn})^{1/n} \text{ bulunur ve}$$

$$f_{alt} = \frac{f_k}{\sqrt{2^{1/n} - 1}} \text{ 'den hesaplanır.}$$

$$\frac{1}{f_{üst}} \approx 1,1 \sqrt{\frac{1}{f_{k1}^2} + \frac{1}{f_{k2}^2} + \dots + \frac{1}{f_{kn}^2}} \text{ 'den bulunur.}$$

6. Geniş bantlı kuvvetlendiriciler:

① RC kompozasyonu (daha çok derbe kuvvetlendiricilerde kullanılır.)



Kompozasyon amaçlı

$$f_c = \frac{1}{2\pi C (r_i || R_g)}$$

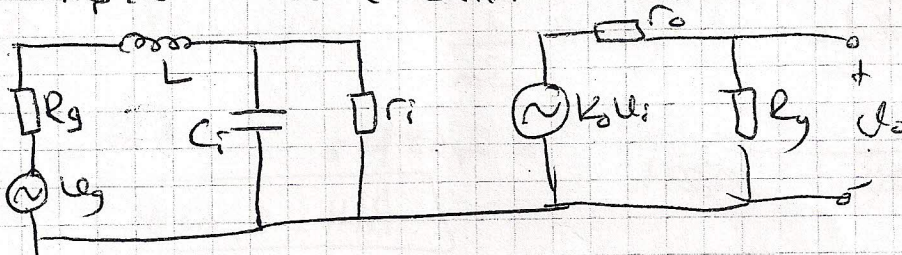
$RC = C \cdot r_i$ seçilerek kompozasyon yapılır;

$$f_c' = \frac{r_i + R + R_g}{2\pi C r_i R_g} \text{ olur.}$$

② CE kompozasyonu:

CE. pF'lar seviyesinde seçilirse yüksek frekansta etkili olur.

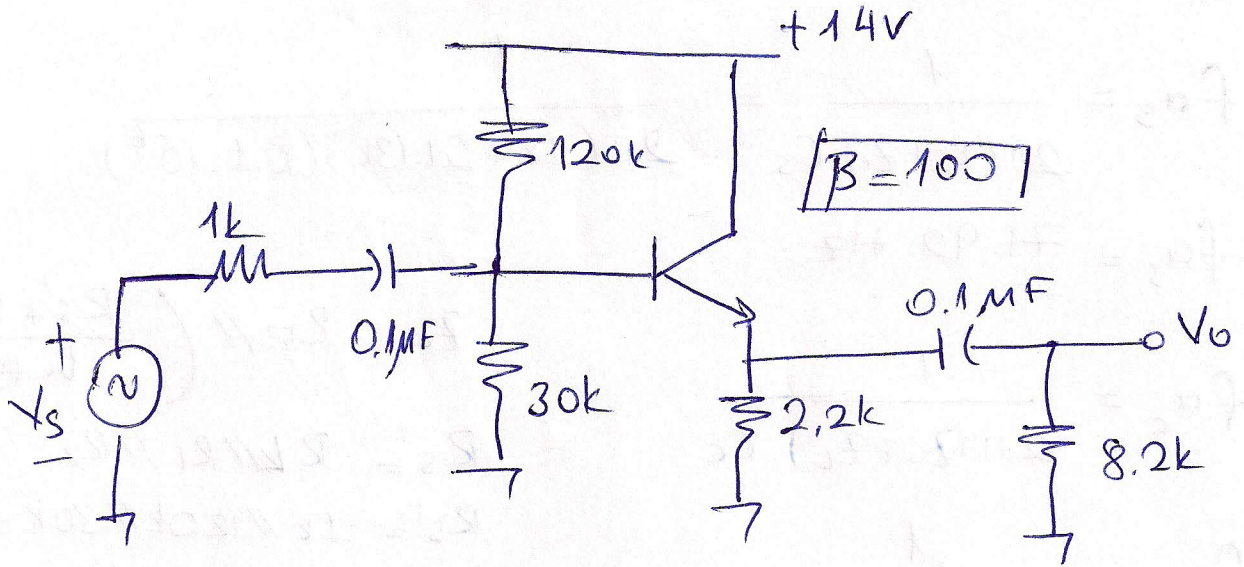
③ RC komp. da kazanç azalır. Bunun yerine LC komp. yapılabilir;



BJT Frekans Cevabi Örnek Soru

(1)

(1)



$$C_{wi} = 8 \text{ pF}, \quad C_{wo} = 10 \text{ pF}$$

$$C_{bc} = 20 \text{ pF}, \quad C_{be} = 30 \text{ pF}, \quad C_{ce} = 12 \text{ pF}$$

$$R_{TH} = 120 \text{ k} \parallel 30 \text{ k} = 24 \text{ k}$$

$$E_{Th} = \frac{30 \text{ k} \cdot (14 \text{ V})}{30 \text{ k} + 120 \text{ k}} = 2.8 \text{ V}$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{TH} + (\beta + 1)R_E} = \frac{2.8 \text{ V} - 0.7 \text{ V}}{24 \text{ k} + (100)(2.2 \text{ k})} \Rightarrow I_B = 8.53 \mu\text{A}$$

$$I_E = (\beta + 1)I_B \Rightarrow I_E = 0.86 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{0.86 \text{ mA}} \Rightarrow r_e = 30.23 \Omega$$

$$h_{ie} = h_{fe} \cdot r_e$$

$$h_{ie} = 100(30.23) \Omega$$

$$h_{ie} \approx 3 \text{ k}$$

Orta Frekans Bölgesinde Kazanç $A_v = \frac{R_E \parallel R_L}{r_e + R_E \parallel R_L}$

$$A_v = \frac{2.2 \text{ k} \parallel 8.2 \text{ k}}{30.23 \Omega + 2.2 \text{ k} \parallel 8.2 \text{ k}} \Rightarrow \boxed{A_v = 0.983}$$

$$Z_i = R_{TH} \parallel \beta(r_e + R_E')$$

$$R_E' = R_E \parallel R_L = 1.735 \text{ k}$$

$$Z_i = 24 \text{ k} \parallel (100)(30.23 + 1735) \Rightarrow \boxed{Z_i = 21.13 \text{ k}}$$

ALT KESİM FREKANSI İÇİN;

$$f_{as} = \frac{1}{2\pi(R_s + Z_i) \cdot C_s} = \frac{1}{2\pi(1k + 21.13k)(0.1 \cdot 10^{-6})}$$

$$f_{as} = 71.92 \text{ Hz}$$

$$f_{ac} = \frac{1}{2\pi(Z_o + Z_L) \cdot C_c}$$

$$f_{ac} = \frac{1}{2\pi(39.12\Omega + 8.2k)(0.1 \cdot 10^{-6})}$$

$$f_{ac} = 193.16 \text{ Hz}$$

$$Z_o = R_E // \left(\frac{R_s' + h_{ie}}{h_{fe}} \right)$$

$$R_s' = R_k // R_1 // R_2$$

$$R_s' = 1k // 120k // 30k = 0.96k$$

$$Z_o = (2.2k) // \left(\frac{0.96k + 100(30.2)}{400} \right)$$

$$Z_o = 39.12\Omega$$

$$f_{alt} = \max(f_{as}; f_{ac}) = \max(71.92 \text{ Hz}; 193.16 \text{ Hz})$$

$$f_{alt} = 193.16 \text{ Hz}$$

ÜST KESİM FREKANSI İÇİN;

$$f_{\beta i} = \frac{1}{2\pi R_{THi} C_i}$$

$$f_{\beta i} = \frac{1}{2\pi(955)(58 \cdot 10^{-12})}$$

$$f_{\beta i} = 2.87 \text{ MHz}$$

$$R_{THi} = R_k // R_1 // R_2 // Z_b$$

$$Z_b = h_{ie} + (h_{fe} + 1)(R_E // R_L)$$

$$Z_b = 3k + (101)(2.2k // 8.2k)$$

$$Z_b = 178.2k$$

$$R_{THi} = 1k // 120k // 30k // 178.2k$$

$$R_{THi} = 955\Omega$$

$$C_i = C_{w_i} + C_{be} + C_{bc}$$

↓
Miller Etkisi Yok!

$$C_i = 8pF + 30pF + 20pF$$

$$C_i = 58pF$$

$$f_{\beta_0} = \frac{1}{2\pi R_{TH2} \cdot C_0}$$

$$R_{TH2} = R_E // R_L // \left(\frac{h_{ie} + R_X // R_{BB}}{h_{fe}} \right) \quad (2)$$

$$f_{\beta_0} = \frac{1}{2\pi \cdot (38.94) (22 \cdot 10^{-12})}$$

$$R_{TH2} = 2.2k // 8.2k // \left(\frac{3k + 1k // 24k}{100} \right)$$

$$R_{TH2} = 38.94 \Omega$$

$$f_{\beta_0} = 185,875 \text{ MHz}$$

$$C_0 = C_{w_0} + C_{ce} = 10 \text{ pF} + 12 \text{ pF}$$

$$C_0 = 22 \text{ pF}$$

$$f_{\beta} = \min(f_{\beta_a}; f_{\beta_0}) = \min(2.87 \text{ MHz}; 185,875 \text{ MHz})$$

$$\boxed{f_{\beta} = 2.87 \text{ MHz}}$$
